

Lecture 10 Position and Momentum

Reading: S&N Secs. 1.5-1.6

We now understand 2-state sys.
And we are pros at Dirac notation.
Now we will look at position and momentum operators:

$\hat{x}|x\rangle = x|x\rangle$ and $\hat{p}|p\rangle = p|p\rangle$
operator number
We know these are incompatible operators:
 $[\hat{x}, \hat{p}] = i\hbar \hat{1}$

For the next lectures we will investigate how this comes about.

Position Eigenkets of $\hat{x}$ are postulated to form a complete basis.

Any state can be represented as:

$$|\alpha\rangle = \int_{-\infty}^{\infty} dx \langle x|\alpha\rangle |x\rangle = \int_{-\infty}^{\infty} dx \underbrace{\psi_{\alpha}(x)}_{\text{or } \alpha(x)} |x\rangle$$

Imagine a measurement at x' :
It can be described by a projector just like our SG filters:

$$\begin{aligned}\hat{P}_{x'} &= |x'\rangle\langle x'| \\ \Rightarrow \hat{P}_{x'}|\alpha\rangle &= \hat{P}_{x'} \int dx \psi_{\alpha}(x) |x\rangle = \int dx \psi_{\alpha}(x) \hat{P}_{x'} |x\rangle \\ &= \int dx \psi_{\alpha}(x) \delta(x-x') |x'\rangle = \psi_{\alpha}(x') |x'\rangle \rightarrow \text{Back units}\end{aligned}$$

This is an infinitely precise position measurement
"The wavefunction collapse"

Measure it at any other position and you get zero.

$$\langle x'' | \psi_{\alpha}(x') | x' \rangle = \delta(x''-x') \psi_{\alpha}(x')$$

$$\begin{aligned}\text{Equivalently: } \hat{P}_{x''} \hat{P}_{x'} &= |x''\rangle\langle x''| x'\rangle\langle x'| \\ &= \delta(x''-x') |x''\rangle\langle x'| \\ &\sim \delta(x''-x') \hat{P}_{x'}\end{aligned}$$

Again "Measurements are projectors"

Now suppose we only measure position in a Vicinity of x ; i.e., $[x-\frac{\Delta x}{2}, x+\frac{\Delta x}{2}]$.

Then: $\hat{P}_{x'} = \int_{-\Delta x/2}^{\Delta x/2} d\lambda |x'+\lambda\rangle \langle x'+\lambda|$

$\lambda = -\Delta x$ $\lambda = 0$ $\lambda = \Delta x$
 $x' - \frac{\Delta x}{2}$ x' $x' + \frac{\Delta x}{2}$

This makes more sense as a projector. Namely, it is dimensionless and:

$$\begin{aligned} \hat{P}_{x'} |\alpha\rangle &= \int_{-\infty}^{\infty} dx \int_{-\Delta x/2}^{\Delta x/2} d\lambda \Psi_{\alpha}(x) |x'+\lambda\rangle \langle x'+\lambda|x\rangle \\ &= \int_{-\Delta x/2}^{\Delta x/2} d\lambda \Psi_{\alpha}(x'+\lambda) |x'+\lambda\rangle \rightarrow \text{Post Measurement State} \end{aligned}$$

What is the probability of finding the state at a given position x'' ?

$$p(x'') = |\langle x'' | \hat{P}_{x'} | \alpha \rangle|^2 = \left| \int d\lambda \Psi_{\alpha}(x'+\lambda) \delta(x'+\lambda-x'') \right|^2$$

if $x'' \in [x' - \Delta x/2, x' + \Delta x/2]$, then $p(x'') = |\Psi_{\alpha}(x'')|^2$

Turns out this is **Probability Density** (it is dimensionful)

Probability = $|\Psi_{\alpha}(x'')|^2 dx''$ (dimensionless)

Recall the dimensions: $[\delta^{(n)}(x)] = L^{-n}$; $[dx] = L$

So by $\langle x | x' \rangle = \delta(x-x') \Rightarrow [|x'\rangle] = L^{-1/2}$

Say $[\alpha] = L^0$; then $[\Psi(x) \equiv \langle x | \alpha \rangle] = L^{-1/2}$

This is what I meant for you to do in class... sorry!

In other words: $[|\Psi(x)|^2] = L^{-1}$
and the notion of probability only makes sense for intervals

$|\Psi(x)|^2 dx$ is dimensionless.

CHECK: Is the state normalized?

$$\langle \alpha | \alpha \rangle = \left(\int dx \underbrace{\langle \alpha | x \rangle}_{(\langle x | \alpha \rangle)^*} \underbrace{\langle x |}_{\langle x | x' \rangle} \right) \left(\int dx' |x'\rangle \langle x' | \alpha \rangle \right) = \int_{-\infty}^{\infty} dx \Psi_{\alpha}^*(x) \Psi_{\alpha}(x) = \int_{-\infty}^{\infty} dx |\Psi|^2$$

Better be 1.

So $\Psi(x) \xrightarrow{x \rightarrow \infty} 0$ for state to be normalizable.

Same story goes in momentum space

$$|\alpha\rangle = \int_{-\infty}^{\infty} dp |p\rangle \langle p | \alpha \rangle = \int dp \phi(p) |p\rangle$$

• BTW $\langle \alpha | \alpha \rangle = 1 \Rightarrow \phi(p) \xrightarrow{p \rightarrow \infty} 0$.

The translation operator

$\hat{T}(\Delta x) |x\rangle = |x+\Delta x\rangle$ n.b. $\hat{T}(\Delta x) |x, \alpha, p\rangle = |x+\Delta x, \alpha, p\rangle$ Other quantum numbers "labels"

What is $\hat{T}(\Delta x)$? Quite complicated in general but we will find it when $\Delta x \rightarrow dx$.

Consider the state $|\alpha\rangle$:

$$\begin{aligned} |\alpha\rangle &= \int dx \psi_\alpha(x) |x\rangle \longrightarrow \hat{f}(dx) |\alpha\rangle = \int dx' f(dx) |x'\rangle \psi_\alpha(x') \\ &= \int dx' \psi_\alpha(x') |x'+dx\rangle \\ &= \int dx' \underbrace{\psi_\alpha(x'-dx)}_{\text{Translated Wavefunction}} |x'\rangle \end{aligned}$$

Note $\langle\alpha|\alpha\rangle = 1$

$$\Rightarrow \langle\alpha| f(dx)^\dagger f(dx) |\alpha\rangle = \int dx' |\psi_\alpha(x'-dx)|^2 = \int dx' |\psi|^2 = 1$$

So $f(dx)^\dagger f(dx) = 1 \rightarrow$ Unitary Operator

Also: $f(dx') f(dx'') \stackrel{!}{=} f(dx'+dx'')$

And $f(dx)^{-1} = f(-dx)$. Prove These.

And $f(dx) \xrightarrow{dx \rightarrow 0} 1$

Unitary Operators $U^\dagger U = 1$

$$U = \sum_{ij} |i\rangle \langle i| U |j\rangle \langle j| = \sum_{ij} u_{ij} |i\rangle \langle j| = \sum_k u_k |u_k\rangle \langle u_k|$$

$$\text{and } U^\dagger = \sum_{ij} u_{ij}^* |j\rangle \langle i| = \sum_k u_k^* |u_k\rangle \langle u_k|$$

where we used the eigenbasis of U in the last step
(Spectral Theorem)

$$\langle u_i | U | u_j \rangle = u_j \langle u_i | u_j \rangle = u_i$$

and so $U^\dagger U = 1 \Rightarrow \langle u_k | U^\dagger U | u_k \rangle = \langle u_k | 1 | u_k \rangle$

$$\Rightarrow u_k^* u_k = 1 \Rightarrow |u_k|^2 = 1 \text{ for every } k$$

• Unit norm eigenvalue $\Rightarrow u_k = e^{i h_k}$ for some real h_k
It is a fact that

Any unitary operator can be written as

$$U = e^{i \hat{H}} \text{ w/ } \hat{H} \text{ hermitian}$$

Why hermitian?

$$\underline{U^\dagger U = 1} \Rightarrow (1 - i\hat{H}^\dagger + \dots)(1 + i\hat{H} + \dots) = (1 + i(\hat{H} - \hat{H}^\dagger) + \dots) \stackrel{!}{=} 1$$

$$\Rightarrow \boxed{\hat{H} - \hat{H}^\dagger = 0} \text{ must cancel order by order.}$$

Unitary Operators preserve norms:

$$\| |a\rangle \| = \langle a | a \rangle$$

$$\| U |a\rangle \| = \langle a | U^\dagger U |a\rangle = \langle a | a \rangle$$

Very useful as a transformation of bases:

$$\{ |a\rangle \} \rightarrow \{ |b\rangle \} \quad \text{w/} \quad |b\rangle = U |a\rangle$$

Take some operator A w/ eigenvalues

$$A |a\rangle = a |a\rangle$$

Then build $\boxed{U A U^{-1}}$ *Unitary Equivalent Observables.*

Clearly

$$U A U^{-1} U |a\rangle = a U |a\rangle$$

$$(U A U^{-1}) |b\rangle = a |b\rangle$$

$|b\rangle$ are eigenkets of $U A U^{-1}$ with the SAME eigenvalues of A .

$B |b\rangle = b |b\rangle$ We will come back to this ∇

Back to translations:

$$f^\dagger(\vec{d}x) f(\vec{d}x) = \mathbb{1} \quad \text{so} \quad f(\vec{d}x) = 1 - i \vec{K} \cdot \vec{d}x$$

with $\vec{K}$ a vector of hermitian operators.

? Prove $f^\dagger(\vec{d}x) f(\vec{d}x) \cong 1$ to first order.

? Prove $f(\vec{d}x'') f(\vec{d}x') = f(\vec{d}x' + \vec{d}x'')$.

Crucial point:

$$\begin{aligned} [\hat{x}, \hat{f}(\vec{d}x)] | \vec{x} \rangle &= \hat{x} \hat{f}(\vec{d}x) | \vec{x} \rangle - \hat{f}(\vec{d}x) \hat{x} | \vec{x} \rangle \\ &= (\vec{x} + \vec{d}x) | \vec{x} + \vec{d}x \rangle - \vec{x} | \vec{x} + \vec{d}x \rangle \end{aligned}$$

higher order

$$[\hat{x}, \hat{f}(\vec{d}x)] | \vec{x} \rangle = \vec{d}x | \vec{x} + \vec{d}x \rangle = \vec{d}x (| \vec{x} \rangle - \vec{K} \cdot \vec{d}x | \vec{x} \rangle)$$

$$\text{Operator identity} \quad [\hat{x}, f(\vec{d}x)] \equiv \vec{d}x \cdot \mathbb{1}$$

Expand $f(\vec{d}x)$:

$$[\hat{x}, -i \vec{K} \cdot \vec{d}x] = -i \left(\hat{x} \vec{K} \cdot \vec{d}x - \vec{K} \cdot \vec{d}x \hat{x} \right) = \vec{d}x \cdot \mathbb{1}$$

$$\text{or} \quad [\hat{x}_i, K_j] dx_j = i dx_i \cdot \mathbb{1} = i \delta_{ij} dx_j \cdot \mathbb{1}$$

$$\Rightarrow [\hat{x}_i, K_j] = i \delta_{ij} \cdot \mathbb{1} \quad \text{Looks familiar?}$$

Call $\hat{K} = \frac{\hat{P}}{\hbar} = \frac{2\pi}{\lambda}$ and we have that translations are generated by the momentum operator.

$$\hat{f}(\vec{dx}) = \mathbb{1} - i \frac{\hat{\vec{p}} \cdot \vec{dx}}{\hbar}$$

$$[\hat{x}_i, \hat{p}_j] = i\hbar \delta_{ij} \mathbb{1}$$

- Generator $\rightarrow$ Technical meaning
- Translation is a unitary transformation
- These make up a continuous Abelian group

For proof, see S&N Sec. 1.4.5.

$$\langle \Delta A^2 \rangle \langle \Delta B^2 \rangle \geq \frac{1}{4} |\langle [A, B] \rangle|^2$$

$$\Rightarrow \langle \Delta x^2 \rangle \langle \Delta p^2 \rangle \geq \hbar^2/4$$

$$\begin{aligned} \hat{f}(\Delta x') |x'\rangle &= |x' + \Delta x'\rangle \\ &= \lim_{N \rightarrow \infty} \left(1 - i \frac{\hat{p}_x \Delta x'}{N\hbar} \right)^N |x'\rangle \\ &= \exp\left(-i \frac{\hat{p}_x \Delta x'}{\hbar}\right) |x'\rangle \end{aligned}$$

For general translations, the limits factorize and the exponents sum.

This relies on

$$[\hat{f}(\Delta y' \hat{y}), \hat{f}(\Delta x' \hat{x})] = \left[\left(1 - i \frac{\hat{p}_y \Delta y'}{\hbar} - \frac{\hat{p}_y^2 (\Delta y')^2}{\hbar^2} + \dots \right), (y \rightarrow x) \right]$$

$$\simeq - \frac{\Delta x' \Delta y'}{\hbar^2} [\hat{p}_y, \hat{p}_x] \quad \text{But translations in } x \text{ and } y \text{ must commute!}$$

Therefore: $[\hat{p}_x, \hat{p}_y] = 0$ since $\Delta x'$ and $\Delta y'$ are arbitrary.

$$\text{What about } \hat{f}(\vec{dx}) |\vec{p}'\rangle = \left(1 - i \frac{\hat{\vec{p}} \cdot \vec{dx}}{\hbar} \right) |\vec{p}'\rangle$$

$|\vec{p}'\rangle$ is an eigenstate of $\hat{f}(\vec{dx})$. **BTW:** $\hat{f}$ is not Hermitian and its eigenvalue turned out to be complex.

Take some general $|\alpha\rangle$ state.
Act $\hat{f}(\Delta x')$ on it with small $\Delta x'$

$$\begin{aligned}\hat{f}(\Delta x')|\alpha\rangle &= \left(1 - i\hat{p}\frac{\Delta x'}{\hbar}\right)|\alpha\rangle \\ &= \int dx' \hat{f}(\Delta x')|x'\rangle \psi_\alpha(x') \\ &= \int dx' \psi_\alpha(x') |x' + \Delta x'\rangle \\ &= \int dx' \psi_\alpha(x' - \Delta x') |x'\rangle \\ &\stackrel{\text{Taylor expand around } x'}{\approx} \int dx' \left[\psi_\alpha(x') - \Delta x' \frac{d\psi_\alpha}{dx'} \bigg|_{x'} \right] |x'\rangle \\ &= |\alpha\rangle - \Delta x' \left(\int dx' |x'\rangle \frac{d\langle x'|}{dx'} \right) |\alpha\rangle\end{aligned}$$

$\hat{O}$

"Diagonal"
in position basis

What are the matrix elements of $\hat{O}$?

$$\langle x''' | \hat{O} | x'' \rangle = \int dx' \langle x''' | x' \rangle \frac{d}{dx'} \langle x' | x'' \rangle = \frac{d}{dx''} \delta(x''' - x'')$$

$$= -\frac{d}{dx''} \delta(x''' - x'') \rightarrow \text{This form is also useful.}$$

$$\Rightarrow \hat{p}|\alpha\rangle = \int dx' |x'\rangle \left(-i\hbar \frac{\partial}{\partial x'} \langle x' | \alpha \rangle \right)$$

or

$$\langle x' | \hat{p} | \alpha \rangle = -i\hbar \frac{\partial}{\partial x'} \langle x' | \alpha \rangle \quad \text{Momentum Operator}$$

In 3D:

$$\begin{aligned}\langle \vec{x}' | \hat{\vec{p}} | \alpha \rangle &= -i\hbar \vec{\nabla}_{\vec{x}'} \langle \vec{x}' | \alpha \rangle \\ &= -i\hbar \vec{\nabla}_{\vec{x}'} \psi_\alpha(\vec{x}')\end{aligned}$$

have also

- $\langle x' | p^n | \alpha \rangle = (-i\hbar)^n \frac{\partial^n}{\partial x'^n} \langle x' | \alpha \rangle$
- $\langle \beta | p^n | \alpha \rangle = \int dx' \psi_\beta^*(x') (-i\hbar)^n \frac{\partial^n}{\partial x'^n} \psi_\alpha(x')$
- $\langle \vec{x}' | \hat{\vec{p}} \cdot \hat{\vec{p}} | \alpha \rangle = (-i\hbar \vec{\nabla}_{\vec{x}'}) (-i\hbar \vec{\nabla}_{\vec{x}'} \psi_\alpha(x')) = -\hbar^2 \nabla_{\vec{x}'}^2 \psi_\alpha(\vec{x}')$

Foreshadow of plane wave:

$$\hat{T} = \frac{\hat{p}^2}{2m} : \text{Kinetic energy. Imagine } \hat{T}|\alpha\rangle = T|\alpha\rangle$$

$$\langle \vec{x}' | \hat{T} | \alpha \rangle = -\frac{\hbar^2}{2m} \nabla^2 \psi_\alpha(\vec{x}') = T \psi_\alpha(\vec{x}') \Rightarrow \left(\nabla^2 + \frac{2mT}{\hbar^2} \right) \psi_\alpha(\vec{x}') = 0$$

$\Rightarrow$ Solution: $\psi_\alpha(\vec{x}) = N e^{i\vec{k} \cdot \vec{x}}$ with $|\vec{k}|^2 = \frac{2mT}{\hbar^2}$

Eigenstate
of Kinetic
energy operator

Now we will find the simplest wavefunction by using the differential form of our momentum operator:

$$\langle x' | \hat{p} | p' \rangle = -i\hbar \frac{\partial}{\partial x'} \langle x' | p' \rangle$$

$$\Rightarrow p' \langle x' | p' \rangle = -i\hbar \frac{\partial}{\partial x'} \langle x' | p' \rangle$$

call $\langle x' | p' \rangle = \psi_{p'}(x')$

$$p' \psi_{p'}(x') = -i\hbar \frac{\partial}{\partial x'} \psi_{p'}(x')$$

Solve 1st order ODE in x'

$$\psi_{p'}(x') = \langle x' | p' \rangle = N \exp\left(\frac{ip'x'}{\hbar}\right)$$

↳ plane wave → fixed momentum.

N fixed by requiring
 $\langle x' | x'' \rangle = \delta(x' - x'')$

$$\langle x' | \int dp' | p' \rangle \langle p' | x'' \rangle = \int dp' \psi_{p'}(x') \psi_{p'}^*(x'')$$

$$= \int dp' N^2 e^{\frac{ip'(x' - x'')}{\hbar}} = N^2 \hbar \int dk e^{ik(x' - x'')} \stackrel{!}{=} \delta(x' - x'')$$

$$\int dk e^{ikx} = 2\pi \delta(x) \rightarrow \text{useful representation of } \delta(x)$$

$$\Rightarrow N = \left(\frac{1}{2\pi\hbar}\right)^{1/2}$$

What is the position of $\psi_{p'}(x')$?

$$\langle \hat{x} \rangle_{p'} = \langle p' | \hat{x} | p' \rangle = \int dx' \langle p' | \hat{x} | x' \rangle \langle x' | p' \rangle$$

$$= \int dx' |\psi_{p'}(x')|^2 x' = \int_{-\infty}^{\infty} dx' \frac{x'}{2\pi\hbar} = \text{indeterminate}$$

What is the momentum of $\psi_{p'}(x')$?

$$\langle \hat{p} \rangle_{p'} = \langle p' | \hat{p} | p' \rangle = p' \delta(p')$$

must be p' .

In 3D:

$$\psi_{\vec{p}'}(\vec{x}') = N e^{i\frac{\vec{p}' \cdot \vec{x}'}{\hbar}} \Rightarrow N = \left(\frac{1}{2\pi\hbar}\right)^{3/2}$$

Reading 1.7.4 + 1.7.5

Gaussian packets

$$\psi(x) = \frac{1}{(\sqrt{\pi}d)^{1/2}} e^{ikx - x^2/2d^2}$$